

Project Title: 555 Timer Based LED Flasher

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Designed using eSim 2.5 / KiCad PCB design workflow

Project Type: Astable NE555 LED Flasher

Main Components: NE555, R1, R2, R3, C1, C2, LED D1 and Battery B1

Verification: DRC completed with 0 errors and 0 warnings

Submitted as a project report covering schematic design, PCB layout, routing, DRC verification, manufacturing outputs and design considerations.

1. Project Overview

The 555 Timer Based LED Flasher is an electronic circuit designed to generate a continuous flashing effect using an NE555 timer IC. The NE555 is configured as an astable multivibrator, allowing it to generate a periodic output signal without an external trigger.

The output of the NE555 controls an LED. The LED turns ON and OFF periodically according to the timing established by the resistor-capacitor network. The circuit is powered from a battery and was converted from a schematic into a PCB layout using an electronic design automation workflow.

The project demonstrates the practical application of the 555 timer, RC timing, LED current limiting, schematic capture, footprint assignment, PCB placement, routing, Design Rule Check (DRC), 3D visualization and generation of Gerber/drill manufacturing files.

Project Objectives

- Generate a repetitive timing waveform using an NE555 timer.
- Flash an LED at a regular interval.
- Design the complete circuit schematic.
- Assign footprints and create a PCB layout.
- Route the PCB and verify it using DRC.
- Generate manufacturing files such as Gerbers and drill files.

2. Circuit Description

The circuit uses X1, an NE555 timer IC, as the main active component. The timer is configured to operate as an astable oscillator. In this configuration, the timing capacitor repeatedly charges and discharges between the internal threshold levels of the 555 timer, causing the output to alternate between HIGH and LOW states.

R1, R2 and C1 form the principal timing network. The charging and discharging of C1 determine the timing of the output waveform. The output drives LED D1 through R3. R3 acts as a current-limiting resistor and protects the LED from excessive current.

C2 is connected to the control-voltage section of the timer and provides additional stabilization against unwanted fluctuations. B1 supplies the DC power required by the circuit. The ground and supply connections complete the electrical return paths.

3. Component Functions

Reference	Component	Function
X1	NE555 Timer	Generates the periodic timing/output waveform.
B1	Battery	Provides the DC power supply.
R1	Resistor	Part of the RC timing network.
R2	Resistor	Part of the timing/charge-discharge path.
C1	Capacitor	Main timing capacitor; its charge/discharge controls the oscillation.
C2	Capacitor	Provides control-voltage stabilization/decoupling.

Reference	Component	Function
D1	LED	Visual indicator that flashes with the timer output.
R3	Resistor	Limits current through the LED.

4. Schematic Design

The following screenshot shows the project schematic containing the NE555 timer, battery, timing resistors and capacitors, LED and current-limiting resistor.

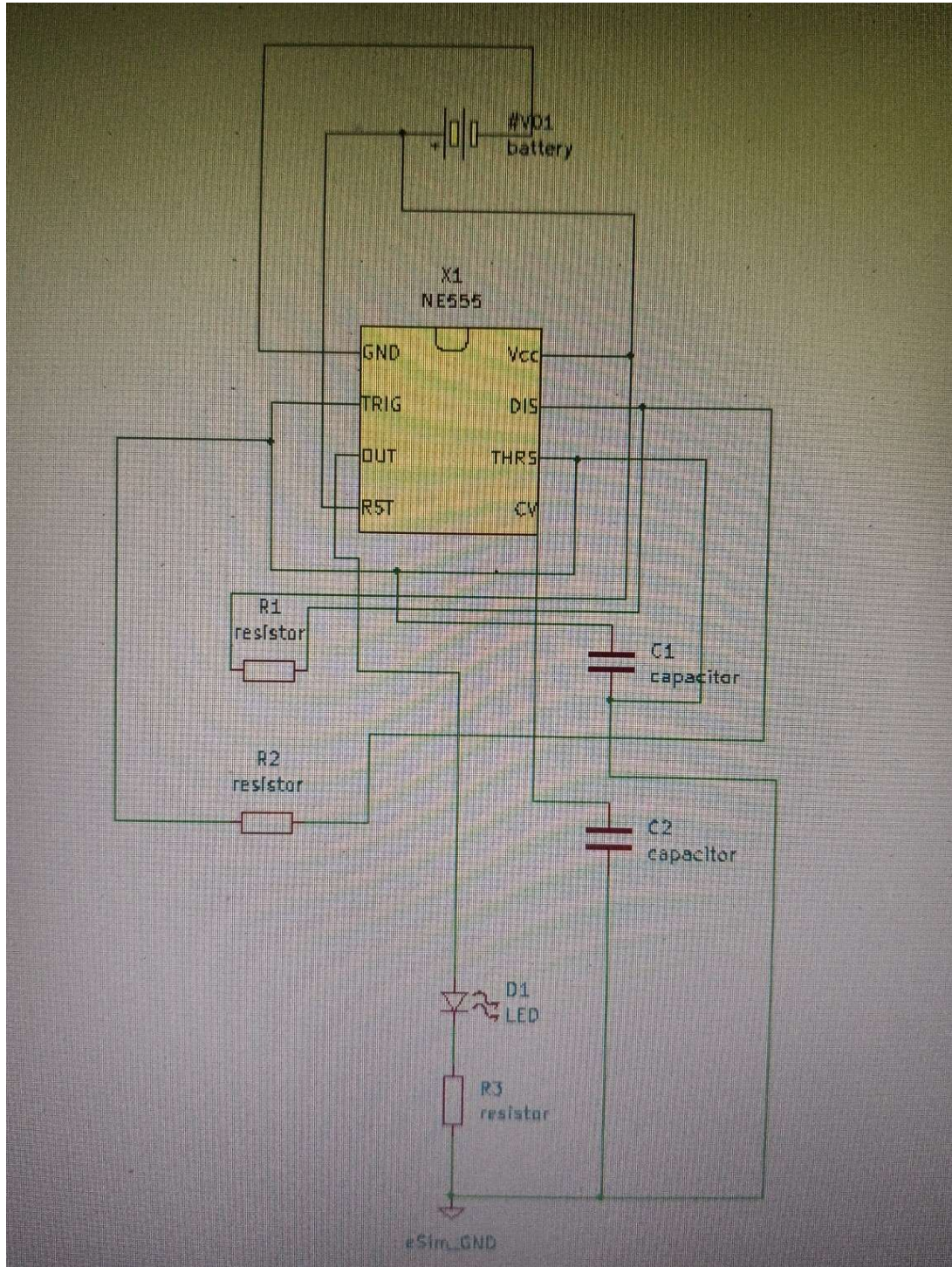


Figure 1. NE555 LED flasher schematic.

5. Working Principle

Power ON: The battery supplies the NE555 circuit. The timing capacitor begins charging through the timing resistor network. When the capacitor voltage reaches the appropriate threshold, the internal 555 circuitry changes state and the output switches.

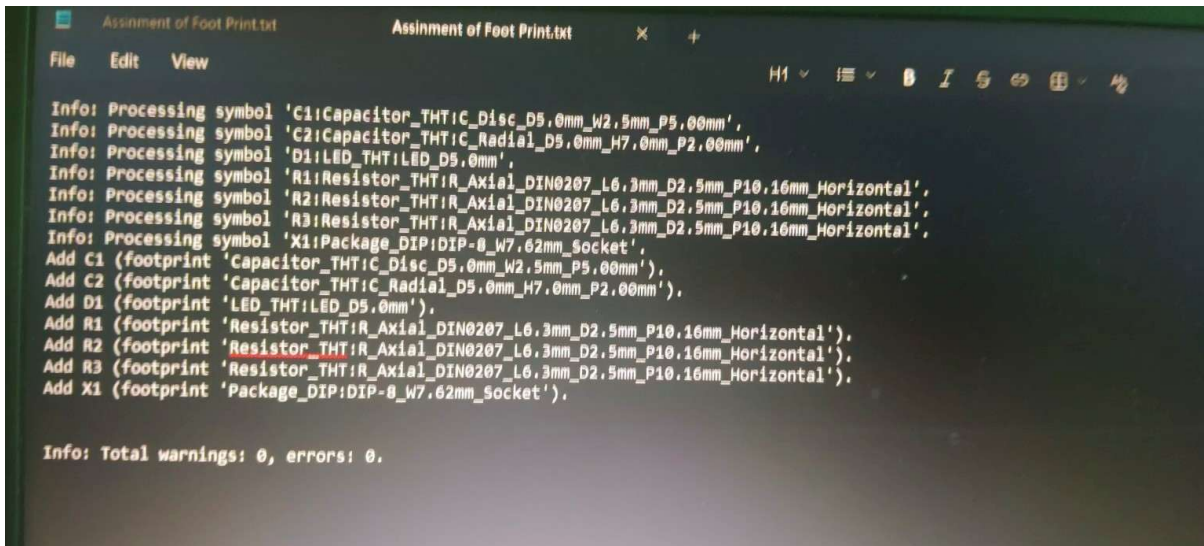
The capacitor then discharges through the timing path. When the voltage reaches the lower threshold, the output switches again. This charge/discharge process repeats continuously. The resulting periodic output drives D1 through R3, so the LED repeatedly turns ON and OFF.

Basic sequence: Battery → NE555 timing circuit → periodic output → LED + current-limiting resistor → flashing LED.

6. Footprint Assignment Details

Footprints were assigned to the schematic components before transferring the design to the PCB editor. The footprint selection should match the physical components available for assembly, including the NE555 package, resistors, capacitors, LED and battery connection.

For the final submission, the exact footprint names should be copied from the project's Footprint Assignment / PCB properties so that the report exactly matches the submitted eSim project. This is preferable to inventing footprint names because the physical package determines hole spacing, pad diameter and board placement.



7. PCB Layout and Routing Methodology

After assigning footprints, the components were transferred to the PCB editor. The components were placed inside the board outline with consideration for component accessibility, short connections and a compact layout. The NE555 was positioned near the center, with timing components and the LED arranged around it.

Routing was performed by connecting the PCB pads according to the electrical netlist. Tracks were arranged to avoid unwanted crossings and to maintain practical connections between the timer, timing network, LED and supply connections. The board outline was defined using the Edge.Cuts layer.

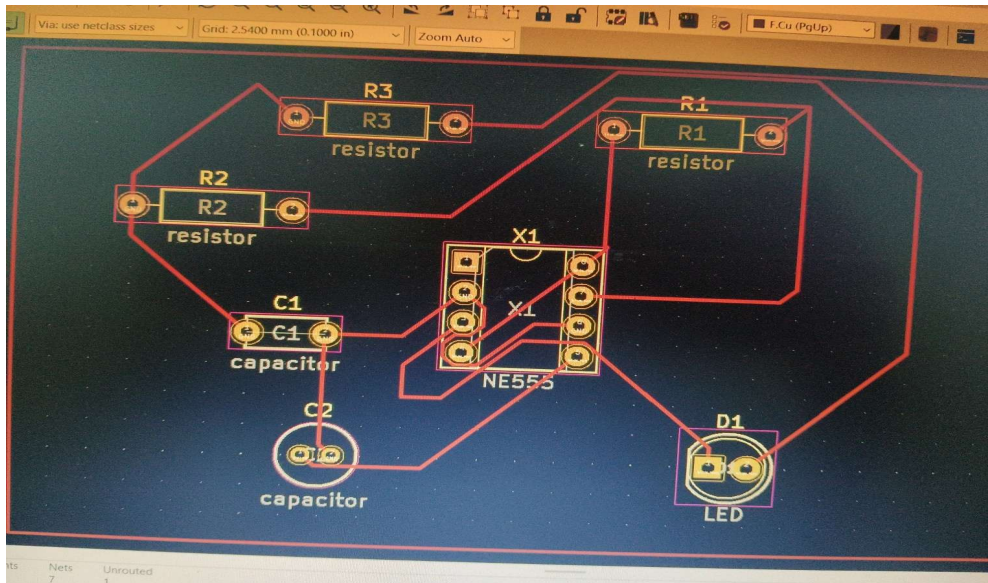
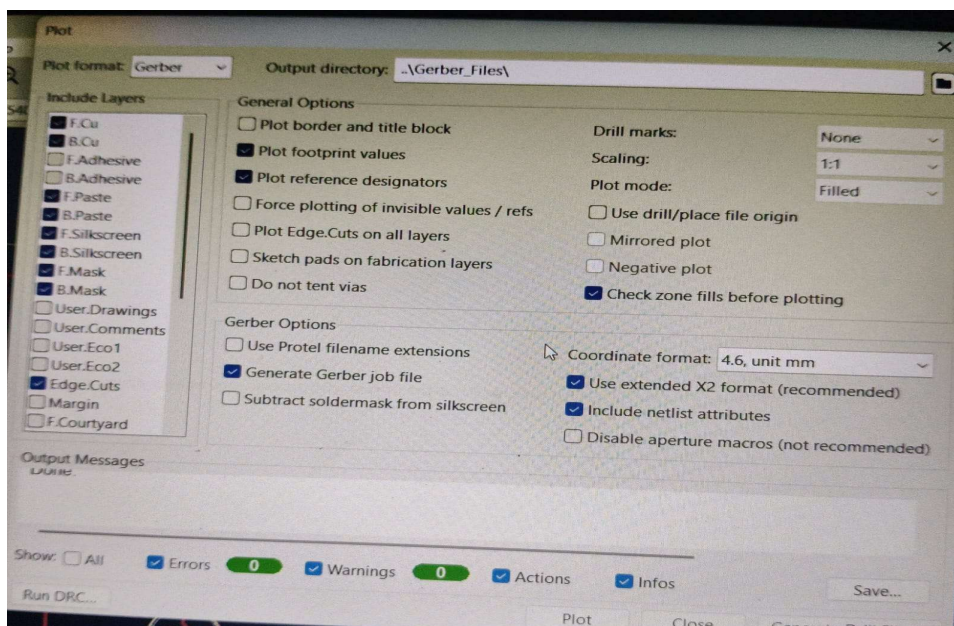


Figure 2. PCB layout and routed tracks.

Design Rule Check (DRC) was performed after PCB routing. The DRC result shown during the design process reported:

DRC Item	Result
Violations / Errors	0
Warnings	0
Unconnected items	0

The DRC result indicates that the completed PCB layout passed the checks that were run, with zero reported errors and zero warnings.



Gerber manufacturing files were generated from the PCB editor. The generated output includes the copper, solder mask, silkscreen and board-outline fabrication layers, together with a Gerber job file. Drill files were also generated for plated and non-plated holes where applicable.

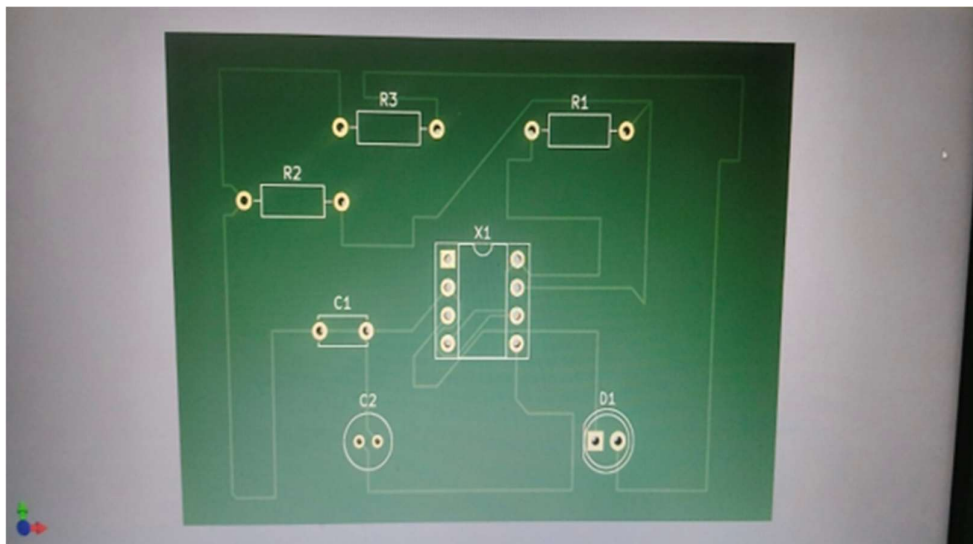
These files are manufacturing outputs and can be supplied to a PCB manufacturer or used with compatible PCB fabrication equipment. The Gerber files do not replace the original eSim/KiCad project files.

Name	Date modified	Type	Size
_autosave-_autosave-esimworkspace-B_C...	12-08-2026 16:17	GBR File	3 KB
_autosave-_autosave-esimworkspace-B_...	12-08-2026 16:17	GBR File	2 KB
_autosave-_autosave-esimworkspace-B_P...	12-08-2026 16:17	GBR File	1 KB
_autosave-_autosave-esimworkspace-B_S...	12-08-2026 16:17	GBR File	1 KB
_autosave-_autosave-esimworkspace-Ed...	12-08-2026 16:17	GBR File	1 KB
_autosave-_autosave-esimworkspace-F_C...	12-08-2026 16:17	GBR File	7 KB
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_autosave-_autosave-esimworkspace-job...	12-08-2026 16:17	GBRJOB File	3 KB
_autosave-_autosave-esimworkspace-NP...	12-08-2026 16:17	DRL File	1 KB
_autosave-_autosave-esimworkspace-PT...	12-08-2026 16:17	DRL File	1 KB

8. 3D Visualization

The PCB should be checked in the KiCad 3D Viewer after footprint assignment and board routing. The 3D view is useful for checking component orientation, physical placement, board outline and approximate mechanical clearances.

Submission note: Add a screenshot from the KiCad 3D Viewer to this section if the final submission requires photographic evidence of the 3D view.



9. Design Considerations

The design was kept compact while maintaining practical component placement and track routing. The LED requires a current-limiting resistor to avoid excessive current. The timing components determine the flashing behavior, so their values should be selected according to the desired frequency and duty cycle.

The PCB outline and routing should be checked for manufacturability. Through-hole pad sizes and hole diameters should match the selected physical components. Power and ground paths should be reliable, and the board should be inspected again after any later modification.

10. Conclusion

The 555 Timer Based LED Flasher project demonstrates the complete process of converting an electronic circuit into a PCB design. The NE555 is used as an astable timer to generate a periodic output that flashes an LED. The schematic was converted into a PCB layout, components were placed, tracks were routed and the design was checked using DRC.

The reported DRC result of 0 errors and 0 warnings indicates that the design passed the performed design-rule checks. Gerber and drill files were generated as manufacturing outputs. The project therefore provides practical experience in circuit design, PCB layout, routing, verification and fabrication preparation.

